A Framework for Benchmarking Machine Learning Methods Using Linear Models for Univariate Time Series Prediction

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***Abstract*—Time series prediction has been attracting interest of researchers due to its increasing importance in decision-making activities in many fields of knowledge. The demand for better accuracy in time series prediction furthered the arising of many machine learning time series prediction methods (MLM). Choosing a suitable method for a particular dataset is a challenge and demands established benchmark methods (BM) for performance assessment. Suppose a particular BM is selected, and an experimental comparison is made with a particular MLM. If the latter does not provide better prediction results for the same dataset, this indicates that some improvements are needed for the MLM. Regarding this matter, adopting a well-established, easy to interpret, and tuned BM is desirable. This paper presents a framework for systematic benchmarking some MLM against well-known Linear Methods (LM), namely Polynomial Regression and models in the ARIMA family, used as BM for univariate time series prediction. We implemented such a framework within the R-Package named TSPred. This implementation was evaluated using a wide number of datasets from past prediction competitions. The results show that fittest LM provided by TSPred are adequate BM for univariate time series predictions.**

I. Introduction

Prediction is knowingly a crucial aspect to decision-making activities. The future states of information about a problem can massively impact on the success or failure of its solution. The time series analysis and its prediction are object of interest of many researchers due to increasing importance and applications in science, business and government [1].

There is a wide variety of models for time series prediction, each one having different properties and complexities, and many of them are state-of-the-art Machine-Learning Time Series Prediction Methods (MLM). Still, none of them is a silver bullet for prediction in time series data. Additionally, time series analysis usually involves the observation of many properties, such as trend and non-stationarity. The existence of these properties leads to the possibility of exploring different data transformations, filters and model fitting methods for obtaining predictions. The number of modeling alternatives and combinations may become very high. Finding an adequate model that solves a time series prediction problem is similar to solving an optimization problem [2]. In such a context, several time series prediction competitions have been organized, including: Santa Fe Time Series Competition [3], EUNITE Competition [4], Competition on Artificial Time Series (CATS) [5], NN3 [6], and NN5 [7] Competitions.

Performance evaluation of prediction models for time series generally involves performing two different and consecutive tasks: (i) training, *i.e.*, finding adjusted parameters that fit a model to a time series given as input; (ii) testing, *i.e.*, predicting subsequent values for the observed time series and comparing them against the actual ones by using an adequate error measure [8]. With this purpose, one usually needs to partition the available time series into two sets, respectively, the training[1] and testing sets. This approach can be used to consistently evaluate the performance of a MLM and appraise its results and errors.

In such scenario, it becomes important to adopt a Benchmark Method (BM) for quality assessment of MLM, *i.e.*, comparing it against other known methods. Given an input time series, values predicted by such BM are computed and compared to the ones found by any MLM. In order to adequately apply this assessment process and secure the validity of its results, the chosen BM should be well-established, reliable and widely adopted. One should also expect the BM to present a prediction performance good enough for creating a demand for MLM refinements. This process iteratively improves the quality of MLM predictions. Ultimately, it leads to improvements in MLM prediction performances, which contributes to advances in the time series prediction field. On the other hand, when no appropriate BM is implemented, the MLM lacks a reference (baseline) to infer its practical value, and also the demand for adjustments in a MLM may go undetected, and necessary improvements in accuracy and methodology may be neglected.

In this paper, we present a systematic framework for benchmarking MLM against well-known Linear Methods (LM), namely Polynomial Regression (PR) and AutoRegressive Integrated Moving Average (ARIMA) models [9], used as BM for univariate time series prediction. Both PR and ARIMA are simple and well-established statistical methods. According to the tuning of their parameters, they are able to produce a family of linear models including the linear, quadratic, autoregressive, random-walk, and moving average models. When

[1]During training, it is a common practice to add a validation phase to measure the quality of fitted model. In such cases, the entire training set is partitioned into actual training and validation sets.

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well-tuned, LM are also able to handle non-stationarity in time series. Furthermore, its predictions are capable of inspiring greater confidence on data scientists since linear models are interpretable, in contrast to some state-of-the-art MLM that are normally black-box models. These characteristics indicate the potential of exploring LM as benchmark for MLM in time series prediction.

The proposed benchmarking framework is implemented within *TSPred* R-Package [10]. The package automatically provides the fittest LM for a given dataset and compares its prediction performance against MLM. The results of such benchmarking process can be useful either for indicating the superiority of MLM, or the need for improvement.

The package was tested with a thorough set of experiments using datasets provided by five past time series prediction competitions. According to our experiments, the fittest LM provided by TSPred served as adequate BM. Not only did they present superior results than a significant set of MLM, but also they ratified the superior performance of the top ranked MLM from the competitions. Should TSPred had properly been used as benchmarking framework, many poor prediction submissions would have been avoided.

The remainder of this paper is organized as follows. Section II discusses fundamentals of the proposed methodology and experiments. Related work is presented in Section III. Section IV gives a description of the devised experimental methodology for assessing the adequacy of the TSPred provided fittest LM as BM for time series prediction. Section V describes the competitions datasets and settings adopted in our experiments, and also presents an analysis of the yielded results. Conclusions and final remarks follow in Section VI.

II. Univariate Time Series and Linear Methods

A. Basic Time Series Background

A time series can be defined as a set of observations on an object of interest collected over time and possibly presenting correlation between adjacent points in time [11]. In this sense, a time series x_t is a sequence of observations $< x_1, \ldots, x_n >$, where x_1 is the first observation and x_n the last one. The length n of a time series x_t is represented by $|x_t|$.

Although a substantial variety of time series models exists, many of them assume the underlying time series is stationary [12]. A stationary time series x_t is a stochastic process such that: (i) the mean function μ_t is constant and does not depend on time t; and (ii) the auto covariance function between x_t and the time shifted time series x_s depend only on the difference between them $|s - t|$ [12]. If x_t violates any of these constraints, then x_t is considered non-stationary.

Under the non-stationary assumption, one may often be concerned about ways to transform x_t into its corresponding stationary time series. Very often a time series x_t presents stationary behavior around a trend. Thus, it is a common approach to address non-stationarity in a given time-series through differentiation. The first difference, denoted as $\nabla x_t = x_t - x_{t-1}$, is capable of eliminating a linear trend. A second difference can eliminate a quadratic trend, and so on. Equation 1 presents the backshift operator, where ∇^d is the d-th differentiation and the operator $(1 - B)^d$ can be expanded for higher levels of differentiation [11].

$$\nabla^d = (1 - B)^d, \quad B^k x_t = x_{t-k} \tag{1}$$

B. Linear Models for Time Series

We can generally consider a method for estimating a time series model to be either *rigid* or *flexible*. Rigid methods, such as linear regression, make initial assumptions about the time series and impose a particular model on it. The modeling process can then be briefly understood as a problem of estimating a set of coefficients, presenting low computational costs. Nevertheless, the adopted model might not well-adjust to the entire time series [13].

In contrast, flexible methods are data-driven and focus on fitting a model such that the available time series observations are approximated as much as possible, given a certain degree of smoothness [13]. Some examples of flexible methods are the regression spline [13], and the ML methods [14], [15].

An important issue regards the interpretability of a time series model. Managers and researchers tend to feel safer when they understand models. The models developed by rigid methods tend to be considerably easier to interpret, hence they might be preferable if the advantage of flexible methods is not statistically significant or just slightly better [1]. Two of the most important rigid methods for time series prediction are polynomial regressions and ARIMA models [12].

C. Polynomial Regression (PR)

Linear Regression (LR) assumes a time series x_t linearly depend on a set of input series, $z_{t1}, z_{t2}, \ldots, z_{tq}$. LR is presented in Equation 2, where $\beta_1, \beta_2, \ldots, \beta_q$ are unknown fixed coefficients, and ϵ_t is the error term [11].

LR can be extended to handle high order relationships through polynomial approximation functions over input series. This approach is known as polynomial regression (PR). Let l be the order of PR. Equation 2 can be coerced into Equation 3 [13], providing the general formulation for PR of order l.

$$x_t = \beta_0 + \beta_1 z_{t1} + \beta_2 z_{t2} + \cdots + \beta_q z_{tq} + \epsilon_t \tag{2}$$

$$x_t = \beta_0 + \beta_1 z_{ti} + \beta_2 {z_{ti}}^2 + \cdots + \beta_l {z_{ti}}^l + \epsilon_t \ (i = 1, \ldots, q) \tag{3}$$

D. ARIMA

ARIMA is derived as a composition of the autoregressive and moving average modeling processes, with the possible addition of a preliminary differentiation process [9]. Autoregressive models (AR) assume that a value x_i of a time series can be described as a function of its p past values, $x_{i-1}, x_{i-2}, \ldots, x_{i-p}$. An autoregressive model of order p, AR(p), is defined in Equation 4, where x_t is stationary, $\phi_1, \phi_2, \ldots, \phi_p$ are constants ($\phi_p \neq 0$), and ω_t is a Gaussian white noise series with mean zero and variance σ^2_ω [9].

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + \omega_t \tag{4}$$

Conversely, the moving average model (MA) assumes that a value x_i of a time series can be described as a function

of its q past white noise values $\omega_{i-1}, \omega_{i-2}, \ldots, \omega_{i-q}$. The moving average model of order q, MA(q), is defined in Equation 5, where q is the number of moving average lags and $\theta_1, \theta_2, \ldots, \theta_q$ ($\theta_q \neq 0$) are parameters [11]. The MA model also differs from AR as it is stationary regardless of the values assumed by the parameters $\theta_1, \ldots, \theta_q$ [11].

$$x_t = \omega_t + \theta_1 \omega_{t-1} + \theta_2 \omega_{t-2} + \cdots + \theta_q \omega_{t-q} \quad (5)$$

Both AR and MA processes can be combined resulting in the autoregressive moving average (ARMA) models. The ARMA(p, q) model is defined in Equation 6, where $\phi(B)$ and $\theta(B)$ are the AR and MA operators, defined in Equations 7 and 8, respectively [9].

$$\phi(B)x_t = \theta(B)\omega_t \quad (6)$$

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p \quad (7)$$

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q \quad (8)$$

Ultimately, the integrated ARMA model, or ARIMA model, can be formulated by including a prior differencing process, therefore devised to handle non-stationary time series. Thus, a time series x_t is defined as an autoregressive integrated moving average model of order (p, d, q), *i.e.*, ARIMA(p, d, q), as presented in Equation 9 [11].

$$\phi(B)(1 - B)^d x_t = \theta(B)\omega_t \quad (9)$$

In many situations, a time series can even seem to behave in a completely random manner over time. In that case, a useful model to estimate such time series is the ARIMA(0, 1, 0), trivially considered a random-walk with drift model [11], in which the value of a time series at time t is basically explained by the value of the series at time $t - 1$, plus a completely random movement represented by ω_t, plus a drift constant δ.

E. Finding Adequate Linear Models

The general approach to produce a Polynomial Regression (PR) model is to start from linear regression. Then, a higher order l in the PR model is progressively introduced. Each time a new order is introduced, ANOVA statistical tests are conducted to measure if the added order provides a significant better model. This steps continue until no improvement is achieved [13].

The general step to fit an ARIMA model is to identify the autoregressive order p, the order of differencing d, and the moving average order q [9]. Choosing the correct parameters to produce a ARIMA model is not a trivial task. Some algorithms have been developed [16], [17] to compute the parameters p, d and q.

III. Related Work

The analysis and time series prediction are search fields widely explored over the past decades. In our research, we were particularly interested in surveying the general adoption of the linear models [9] as benchmark for the assessment of univariate time series prediction methods, especially when challenging state-of-the-art MLM.

Several of the examined papers compare the prediction accuracy of neural network methods with ARIMA models. Jha and Sinha [18] and Yao and Tan [19] performed their comparison in the Sale Forecasting domain; Ho et al. [20] and Ghiassi et al. [21], in the Business or Industrial Applications domain; Zhang [22], in the Stock Market Forecasting domain; Prybutok et al. [23] and Zhang [22], in the Chemical Concentration Forecasting domain; Ghiassi et al. [21], in the Biological Events Forecasting domain; and Zhang et al. [24] used simulated data for the comparison. Many other papers that assess the accuracy of neural networks through ARIMA are surveyed in Zhang et al. [25].

There are also some works that use the ARIMA models for developing hybrid neural network methods, which generally outperform both ARIMA and usual neural networks applied separately. Among them are the works from Khashei and Bijari [26] and Taskaya-Temizel and Ahmad [27]. Although in these cases the ARIMA models are part of the hybrid time series prediction methods, they may still be used as benchmark in a comparison study whose purpose is to infer whether the hybrid approaches are satisfactory.

The ARIMA modeling process is not usually a straightforward task. An improper model adjustment leads to poor predictions with biased and unreliable performance assessment of the time series prediction methods to which it may be compared. Among the examined papers, only Zhang [22] and Zhang et al. [24] used packages to automate the ARIMA modeling task.

Another important issue we have researched is the study and development of benchmarking frameworks and tools for MLM performance assessment in literature. In fact, we have observed that several authors focused on the task of benchmarking different models in many different domains. Ramey [28] and Lessmann et al. [29] developed frameworks for benchmarking classification models and algorithms. Moreover, Bischl et al. [30] and Eugster [31] developed the R-packages *mlr* and *benchmark*, respectively, which provide tools for executing automated experiments of benchmarking a set of methods for data mining tasks such as classification and regression. These packages are mostly designed to support the use of data from relational models and focus on benchmarking based on plot visualization.

Hyndman [16] and Hyndman et al. [32] present frameworks for automatic forecasting using linear methods such as ARIMA and exponential smoothing. Hyndman [16] produced the well-known R-package named *forecast*, which can be used for automatic time series prediction. Furthermore, we note the works of Diebold and Mariano [33], that proposes a variety of tests to compare the predictive accuracy of two different prediction methods, Diebold and Lopez [34] that proposes an ensemble approach with the combination of different prediction methods, and Kumar et al. [35] which propose a class of analytics systems to manage model selection using key ideas from data management research. Moreover, many of the aforementioned techniques are complementary to our approach, as they can be used in one or more activities presented in our

benchmark-based time series prediction process.

All in all, there are many time series prediction works in the literature that conduct ad-hoc comparison and performance assessment using linear models. There are also several works that present benchmarking frameworks and tools for MLM performance assessment. Nonetheless, to the best of our knowledge, there is no other work that proposes and implements a systematic framework for benchmarking MLM for univariate time series prediction using an automatically provided fittest LM to serve as an adequate BM for performance assessment.

IV. Methodology

In this section, we present our methodology for testing our implemented benchmarking framework for performance assessment of MLM against an automatically provided fittest LM (PR or one of the models in the ARIMA family). Both the framework and experimental tests were performed using R, which is both a language and environment with many statistical and graphical packages [36]. We have produced several R functions, which are part of the R-package named *TSPred* [10] for benchmark-based time series prediction available from *http://sourceforge.net/p/gpca/wiki/TSPred/*.

As an introduction to this methodology, a general benchmark-based comparison process for univariate time series prediction is presented in Figure 1. It provides a systematic way of comparing a MLM and a BM for time series prediction that can be used for evaluation[2]. The goal is to use such BM in aiding the analysis of MLM yielded prediction errors and improving the reliability of results analysis. Based on the analysis and comparison of the performances of a MLM against its BM, one may consider refining and perfecting the former to obtain a more accurate model and lower prediction errors. Regarding this task, it is useful to adopt an easy to interpret BM, since the analysis of the way in which the BM is fitted may aid the configuration of the refined MLM. This process may be repeated if the analyzed time series prediction method does not reach a desired performance.

The first activity depicted in Figure 1 (labeled 0) refers to the choice of MLM. Activity 1 refers to acquiring the training and evaluation (*i.e.*, either validation or testing) datasets, and possibly doing some data preprocessing. Activities 2-4 and 5-7 are parallel. The first pair (2 and 5) respectively refers to training the prediction models using a BM and MLM. The next two activities (3 and 6) refer to applying the previously fitted models to an evaluation set. The last pair (4 and 7) corresponds to the evaluation of prediction errors yielded by the models. Finally, activity 8 is the analysis of prediction errors and comparison of performance of MLM and BM. If the results of MLM are not adequate, MLM can be refined, as indicated in activity 9. This entire process may be repeated if the performance of the MLM is not considered adequate.

Henceforth we are particularly interested in testing the adequacy of the fittest LM, automatically provided by TSPred, to

[2]We refer to evaluation as either validation or testing phase according to the target study.

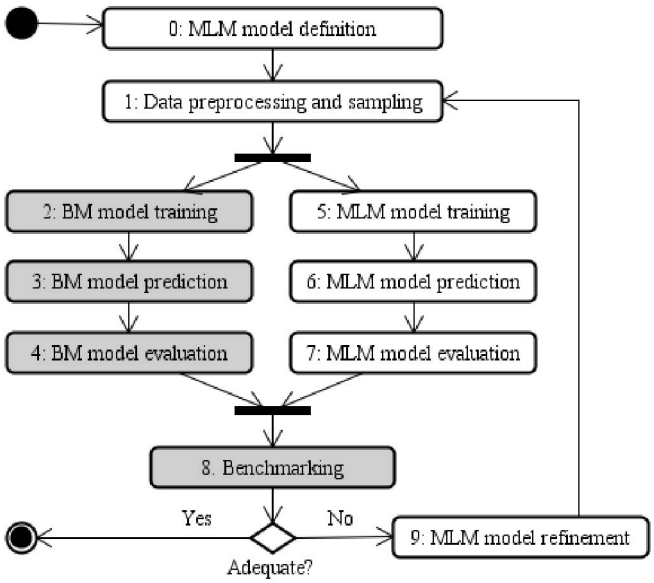


Fig. 1: Activity diagram for benchmark-based time series prediction

the role of BM for benchmarking MLM. With this purpose, we instantiate the presented benchmark-based comparison process using ARIMA models and PR as candidate BM. According to Figure 1, the instantiation performed in our experiments encompasses using such LM in three activities corresponding to (i) model training, (ii) multistep-ahead prediction, and (iii) evaluating prediction error and analysis. Both BM and MLM use exact same training and evaluation sets, which enables us to compare the two results and measure the adequacy of the MLM, i.e., benchmarking the MLM against the BM.

Algorithm 1 presents the EvalFittedLM procedure, which summarizes our experimental methodology for assessing the adequacy of the fittest LM provided by TSPred as BM for time series prediction. Our methodology adopts the use of MLM benchmark results from time series prediction competitions, which provide the products of a wide variety of experimental instantiations of the activities 5-7 of Figure 1. Hence, our EvalFittedLM procedure need only to instantiate the activities depicted in gray in Figure 1. A similar experimental approach was adopted by Hyndman et al. [32]. EvalFittedLM initially receives four parameters: (i) a time series dataset x_t, (ii) a number m of observations to be predicted, (iii) a set of benchmark linear models that is referred as α, and (iv) a set of evaluated prediction errors, ϵ_β, yielded by a set of MLM methods, β, to be evaluated against our proposed BM.

In lines 2 and 3, EvalFittedLM divides x_t into two disjoint subsets, $x_t^{evaluate}$ and x_t^{train}, corresponding to the previously described evaluating and training sets, respectively. The former subset needs to have no more than m observations to enable the subsequent calculation of the prediction errors, therefore we assign the length of $x_t^{evaluate}$ equals to m, and for x_t^{train} we assign all the remaining observations contained in x_t, that

Algorithm 1 Experimental Methodology

Input:

x_t = experimental time series data
m = number of observations to be predicted
α = set of linear models
ϵ_β = prediction errors yielded by a set of ML models

Output:

R = ranking of prediction performances

1: **procedure** EVALFITTEDLM($x_t, m, \alpha, \epsilon_\beta$)
2: $x_t^{evaluate} \leftarrow$ DataSampling(x_t, m)
3: $x_t^{train} \leftarrow x_t - x_t^{evaluate}$
4: $\hat{\alpha} \leftarrow$ TrainModel(α, x_t^{train})
5: $\rho_\alpha \leftarrow$ Predict($\hat{\alpha}, x_t^{train}, m$)
6: $\epsilon_\alpha \leftarrow$ Error($\rho_\alpha, x_t^{evaluate}$)
7: $\epsilon_A \leftarrow$ FittestError($\hat{\alpha}, \rho_\alpha, \epsilon_\alpha$)
8: $R \leftarrow$ Benchmark($\epsilon_A, \epsilon_\beta$)
9: **return** R

is, the length of x_t minus the m observations.

From lines 4 to 6, the algorithm performs training, predicting and error measurement. The first executed task, in line 4, is the tuning of the parameters of the set of linear models, or in this case, α, based on the data within x_t^{train}, which gives us the set of tuned LM models $\hat{\alpha}$ (including PR and one of the models in the ARIMA family). We can then use $\hat{\alpha}$ to predict the m next observations of the time series present in x_t^{train}. The set of predictions are then represented by ρ_α. This task can be observed in line 5 of Algorithm 1. Line 6 computes the prediction errors of the tuned linear models using $x_t^{evaluate}$. Let ϵ_α be the set of prediction errors produced by α.

In line 7 the fittest LM (either PR or one of the models in the ARIMA family), A, is selected and its evaluated prediction error is represented by ϵ_A. Next, in line 8, the evaluated prediction errors ϵ_A and ϵ_β are used to benchmark the set of selected MLM β against the fittest LM A generating a performance ranking referenced as R. Finally, the algorithm returns R for further analysis.

V. Experimental Evaluation

A. Datasets

To help assess the adequacy of the TSPred provided fittest LM as benchmark for time series prediction, we have selected datasets of five time series prediction competitions: Santa Fe [3], EUNITE [4], CATS [5], NN3 [6], and NN5 [7]. These competitions were adopted as they maintain their datasets and results available. In addition, they make accessible the papers describing the applied methods of many competitors. The works presented in them comprehend a large variety of MLM, which indicate great efforts done on the time series prediction problem through years. Furthermore, all the selected competitions provide easy access to the performance metrics used, as well as the ranked prediction errors found by each of their 125 competitors. This enabled us to compare the prediction errors and performance of the competitors' methods against the proposed BM. Thus, the evaluation and analysis of prediction results of MLM was facilitated.

The main dataset of the Santa Fe Competition is the nonlinear and stationary time series A with 1,000 observations. We have also used the computer-generated, nonlinear and non-stationary time series D, containing 100,000 observations.

The EUNITE Competition was interested in solving the problem of predicting the maximum daily electrical loads for January 1999 based mainly on real values of half-an-hour loads and average daily temperatures measured between 1997 and 1998. This dataset has over 35,000 data points.

The CATS Competition presented an artificial time series with 5,000 observations, among which 100 are unknown and grouped into five non-consecutive gaps of 20 successive values. The dataset allowed for using imputation techniques [1], which cannot be applied for the fifth set of unknown values.

The NN3 Competition was devised with a dataset composed of 111 monthly time series drawn from homogeneous population of real empirical business time series. Each of these time series possess from 50 to 126 observations.

The NN5 Competition extends the earlier NN3 competition offering a dataset with 111 time series originated from daily withdrawals at 111 different cash machines within England. Each of these time series possesses 735 observations, and may present missing data.

B. Experimental Settings

In a conventional scenario, in order to fit a model to a given time series x_t, first we would have to partition x_t into training and evaluating sets. However, since our chosen prediction competitions are finished, both their time series datasets and their results were available for use in our experiments. Hence, we used the entire provided time series as training sets and the results were used as testing sets. This allowed us to compare linear models with each MLM used in the competitions. Additionally, although TSPred explores both type of linear models (PR and models in the ARIMA family) and is able to return the fittest linear model among them as the benchmark linear model, for the sake of comparison, we provided in our experiments results for both fittest PR and ARIMA models. During PR model fitting, we have built PR models up to the fifth order for each training set. The order of each fittest PR model was selected based on the systematic analysis of ANOVA statistical tests, and their coefficients were estimated using the R function *lm*.

In ARIMA model fitting, we have used the *auto.arima* function of the *forecast* R-package [16] to automate the tuning of the ARIMA models. The *auto.arima* function selects an optimized ARIMA model defined by the values of its parameters, p, d and q, computed using a variation of the Hyndman and Khandakar algorithm [16], [17]. The models evaluated by

auto.arima include AR, MA, ARMA and ARIMA, besides random-walk and white noise with drift, *i.e.*, ARIMA(0,1,0) and ARIMA(0,0,0), respectively.

After training LM models, we used them to predict the next observations of the modeled time series as required by each competition. We have measured the prediction errors over the selected testing sets for each experiment using the error metrics defined by their respective competitions[3].

C. Analysis of the Linear Models Overall Performance

During our experiments, we have trained linear models (PR and models in the ARIMA family) over 235 different time series. We studied the performance of such linear models as benchmark methods and also conducted a relative comparison between PR and ARIMA models. The diversity of the time series used makes our analysis less biased and helped us to come up with overall conclusions about our proposed BM performance. We can observe from Table III that the overall performance of the ARIMA models was considerably superior to PR models. However, PR was better suited for the EUNITE dataset.

The systematic tuning of ARIMA parameters allowed us to estimate the time series using a variety of linear models from ARIMA family. Table I presents the ARIMA models produced in our experiments, where most selected models were in fact ARMA and ARIMA.

TABLE I: List of ARIMA models produced by the experiment

Model	AR	MA	ARMA	ARIMA	Random Walk	White Noise	Total
Qt.	18	17	142	26	1	35	239
%	8	7	59	11	0	15	100

Besides prediction errors, our experiments provided plots depicting ARIMA models predictions and their confidence intervals along with the actual data presented on the testing sets. These graphics helped us understand our results. An example is presented in Figure 2, where we can easily notice the progressive stretch of the confidence interval and loss of representativeness of the ARMA model predictions. This is common in multistage prediction, in which errors committed in the past are propagated into future predictions [37].

The computed NMSE errors yielded from the ARIMA models for the time series of the Santa Fe Competition can be observed in Table III. We can see that the ARIMA predictions were reasonable, given that the used time series were highly nonlinear. Particularly, we can see that the ARIMA prediction performance was comparatively better for time series D, which is 100 times greater than time series A, thus possessing more data to help training the ARIMA models.

The results for the EUNITE competition are also shown in Table III. The problem posed by the EUNITE Competition was specially challenging as the errors of the average daily temperatures predictions are propagated to the predictions of the daily maximum electrical loads, which was our main concern. Nonetheless, according to Table III, PR results were also reasonable and outperformed the majority of results of the competitors.

[3]For further description and R code applied to each adopted competition, please refer to our web page [10].

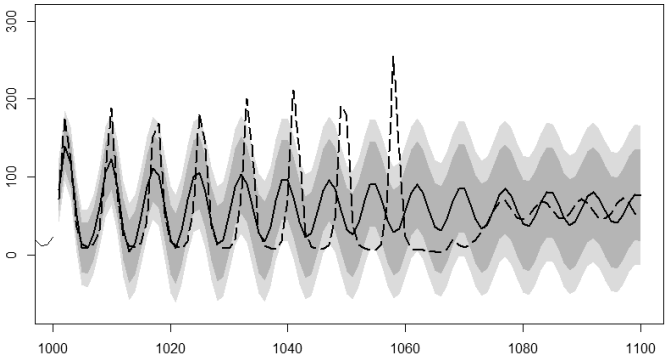


Fig. 2: ARMA predictions (solid line) for the time series A of the Santa Fe Competition. The actual time series values are represented by the dashed line.

TABLE II: ARIMA models' MSE prediction errors for each gap of unknown values of the CATS time series

	Gap 1	Gap 2	Gap 3	Gap 4	Gap 5
MSE errors	575.787	531.547	1,978.816	583.143	2,196.485

The prediction errors for ARIMA models computed for the CATS Competition dataset are presented in Table II, where MSE errors for each one of its five gaps of unknown values can be observed. We applied interpolation techniques (forward and backward ARIMA forecast) to predict the first four gaps. The prediction errors of these gaps are much inferior when compared to the one of the fifth gap, for whose prediction we could not have applied the same interpolation techniques. Hence, we can conclude that the improvement in the performance of the ARIMA models when using interpolation procedures was coherent with previous knowledge of multi-stage prediction approaches [37].

Among the 235 time series used to train ARIMA and PR models, 222 were obtained from the NN3 and NN5 datasets. The box plot graphics of the SMAPE prediction errors for the time series in the NN3 and NN5 datasets, depicted in Figure 3.a and Figure 3.b, respectively, indicate a general consistency in the performance of the trained ARIMA models (fittest LM) over these competitions. Furthermore, the analysis of these graphics indicates that the mean SMAPE prediction errors computed for these two competitions, observed in their respective rankings in Table III, are actually quite representative. This fact reassures the validity and reliability of the comparison of our BM's prediction errors to the ones from methods applied by these two competitions' participants in their rankings.

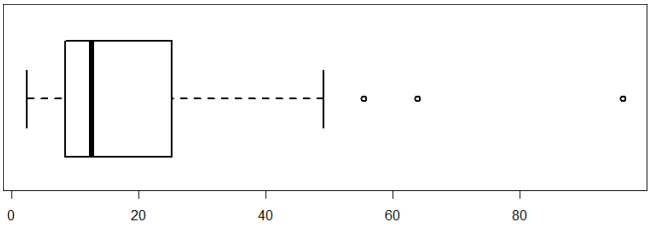
There are two main characteristics one seeks on a satisfactory BM. Firstly, it should be capable of providing a minimum acceptable level of prediction performance and therefore en-

TABLE III: Rankings of the top 25 results of the chosen competition datasets including results from TSPred R-package

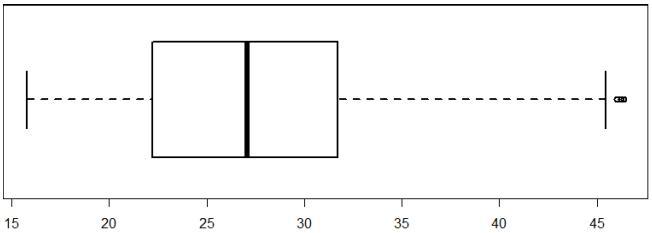
Rank	Santa Fe				EUNITE		CATS			NN3		NN5	
	Dataset A		Dataset D							Dataset A		Dataset A	
	index	NMSE	index	NMSE[1]	Participant	MAPE [%]	Participant	E1	E2	Participant	Mean SMAPE	Participant	Mean SMAPE
1	W	0.02	ZH	0.08	Chih-Jen Lin	1.982	Sarkka*	408	346	Illies*	15.18%	Andrawis	20.40%
2	Sa	0.08	**TSPred$_{(ARIMA)}$**	**0.54**	Esp	2.149	Cai*	441	402	Adeodato*	16.17%	Vogel	20.50%
3	M	0.38	U	1.30	Brockmann	2.498	Kurogi*	502	418	Flores*	16.31%	D'yakonov	20.60%
4	L	0.45	**TSPred$_{(PR)}$**	**1.61**	**TSPred$_{(PR)}$**	**2.779**	Hu*	530	370	Chen*	16.55%	Rauch	21.70%
5	U	0.62	Z	4.80	Zivcak	2.873	Palacios-Gonzalez	577	395	D'yakonov	16.57%	Luna	21.80%
6	A	0.71	C	6.40	Kowalczyk	2.985	Maldonado*	644	542	Kamel*	16.92%	Wichard	22.10%
7	McL	0.77	W	7.10	Lewandowski	3.223	Simon*	653	351	Abou-Nasr	17.54%	Gao	22.30%
8	**TSPred$_{(ARIMA)}$**	**0.90**	S	17.00	Kowalczyk	3.264	Verdes*	660	442	Theodosiou*	17.55%	Puma-Villanueva	23.70%
9	**TSPred$_{(PR)}$**	**0.99**			Ortega	3.380	Chan*	676	677	**TSPred$_{(ARIMA)}$**	**17.79%**	Dang	25.30%
10	N	1.00			King	3.388	Wichard*	725	222	de Vos	18.24%	Pasero	25.30%
11	P	1.30			Lotfi	3.389	Beliaev*	928	762	Yan	18.58%	Adeodato	25.30%
12	Can	1.40			Guijarro	3.421	Kong	954	994	C49	18.72%	undisclosed	26.80%
13	K	1.50			Weizenegger	3.694	Wang	1037	402	Perfilieva*	18.81%	undisclosed	27.30%
14	Sw	1.50			**TSPred$_{(ARIMA)}$**	**3.820**	Cellier*	1050	278	Kurogi*	19.00%	**TSPred$_{(ARIMA)}$**	**27.80%**
15	Y	1.50			Boger	3.958	Crone*	1156	995	Beadle	19.14%	Tung	28.10%
16	Car	1.90			Bontempi	3.997	**TSPred$_{(ARIMA)}$**	**1173**	**917**	Lewicke	19.17%	undisclosed	33.10%
17					Pelikan	4.348	Acernese*	1247	1229	Sorjamaa*	19.60%	undisclosed	36.30%
18					Brockmann	4.373	Yen-Ping*	1425	894	Isa	20.00%	undisclosed	41.30%
19					Pelikan	4.437	**TSPred$_{(PR)}$**	**7387**	**6778**	C28	20.54%	**TSPred$_{(PR)}$**	**41.50%**
20					Rivieccio	4.502				Duclos-Gosselin	20.85%	undisclosed	45.40%
21					Brockmann	4.580				Papadaki*	22.70%	undisclosed	53.50%
22					Ivakhnenko	4.653				Hazarika	23.72%		
23					Brockmann	4.712				C17	24.09%		
24					Brockmann	5.087				Njimi*	24.90%		
25					Brockmann	5.425				Pucheta*	25.13%		

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[1] NMSE error for the 15 first predicted observations



(a) Prediction errors of the NN3 Competition dataset



(b) Prediction errors of the NN5 Competition dataset

Fig. 3: Box Plot of ARIMA's SMAPE prediction errors for all time series in the NN3/NN5 Competition dataset (Dataset A)

able the evaluation of the practical viability and expediency of a prediction method. Additionally, a reliable benchmark should be able to offer a way to expose the merit of a time series prediction method that presents higher performance.

According to our experimental results, the fittest LM provided by the package *TSPred* fulfill the role of an adequate BM. Table III presents prediction errors rankings for the top 25 methods in all competitions. It is possible to observe that TSPred's fittest LM maintained an average rank. MLM presenting performance below the fittest LM clearly demand further refinement. Furthermore, prediction errors of the tuned fittest LM models ratified the importance of results from the top ranked methods, which were significantly superior.

VI. Conclusion

Although the importance of using a benchmark process to help in the evaluation of time series prediction methods is well known, it is often neglected. To address this problem, this paper presents a benchmarking framework to ease the process of evaluating the performance of MLM against a fittest LM during time series prediction modeling. The framework was developed within the R-package named TSPred to automate the nontrivial task of tuning and selecting a fittest LM, among PR and models in the ARIMA family, which is then used as benchmark for MLM. Thus, we minimize the risk of improper tuning of LM, and consequently, biased performance assessment of MLM to which it is compared.

We have performed a set of experiments using datasets selected from past time series prediction competitions, allowing the study of the fittest LM provided by TSPred (used as BM)

among a wide range of MLM. Our experiments show the strength of the BM provided by TSPred when compared to a considerable number of MLM. Furthermore, the top ranked MLM significantly outperform the provided BM, therefore certifying their effectiveness. These results indicate the advantages of using the benchmarking framework implemented in TSPred for time series prediction during training and testing.

As future work, we intend to conduct comparative studies of novel MLM against benchmark LM using TSPred. In particular, our next step is to assess novel techniques for data normalization [8], [38] to improve accuracy and efficiency of prediction algorithms in non-stationary time series. Experimental results should be generated using neural networks and SVM to forecast macroeconomic and scientific data.

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